Functional requirements

Accounts and website

- Visitors can view the home page and membership plans, then sign up with email and password, and upload a profile photo.
- Email is verified before the account is active.
- Users log in by role: admin, reception, trainer, or member. Each role sees only what it is allowed to.
- The website offers a mobile app download.

Members and plans

- The system keeps member profiles with contact details, goals, health notes, and a photo.
- Only the admin creates plans and sets prices, and the system allows one active membership per member at a time.
- Admin and reception can search members and assign a trainer.

Payments and expiry

- Members pay online (Chapa), and the membership activates only after the payment provider confirms it.
- Reception can record cash and bank transfer payments.
- Memberships expire automatically, with email reminders 7 days and 1 day before.
- Members can see payment history, and the admin can see revenue.

Check-in and attendance

- The member shows a QR code that refreshes every 60 seconds, and reception scans it.
- The scan shows name, photo, plan, payment, expiry, and trainer, then allows or denies entry.
- Reception can check members in manually by search.
- The admin can optionally turn on self check-in, where the member scans a gym QR code.
- All check-ins are logged, and a repeat check-in within 2 hours is ignored.

Walk-in registration

- Reception registers a new person (name, phone, email, photo), picks a plan, records the payment, and assigns a trainer.
- The new member gets an email invitation to set a password.

Trainers and workout plans

- The admin manages trainers.
- Trainers see their members and create, edit, and reuse workout plans from an exercise library.

- Members view their plan and mark workouts done.

Progress and goals

- Members set goals and log weight and body measurements.
- The dashboard shows a progress list and graphs, and trainers can view and comment.

Classes, bookings, and notifications

- Admin and trainers create classes, and members book places.
- Full classes have a waitlist, and members can cancel their bookings.
- Notifications go out by in-app message, push, and email, including expiry, payment, class, and announcement messages.

Equipment and maintenance

- The admin tracks equipment status, logs maintenance, and gets alerts when service is due.
- Staff can report faulty equipment.

AI assistant (Gemini)

- The assistant drafts workout plans, which a trainer approves before the member sees them.
- It explains exercises, summarizes progress, and answers fitness questions.
- Every answer carries a “not medical advice” notice, and requests are limited per user.

Dashboards and reports

- Member, reception, trainer, and admin each have their own dashboard.
- The admin can export members, payments, and attendance reports.

Non-functional requirements

- **Security:** all traffic uses HTTPS, passwords are stored hashed, and access depends on role.
- **Privacy:** health notes and progress data are visible only to the member, their trainer, and the admin.
- **Performance:** a check-in result appears within 2 seconds.
- **Availability and backup:** the system runs on OCI, with daily database backups.
- **Usability:** the website works on desktop and phone browsers, and the mobile app works on common Android versions.
- **Reliability of payments:** a payment is never processed twice.
- **Auditability:** payments, check-ins, plan changes, and price changes are logged with the user and time.
- **AI safety:** the Gemini key stays on the server and is never exposed to the website or the app.